LOGARITHMIC, HIGH-DENSITY FIBRE, AND FINITE-FIBRE OBSTRUCTIONS TO INFINITE PRODUCT CONFIGURATIONS

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Abstract. We study several infinite product-type configurations and partition refinements appearing in questions of Kra, Moreira, Richter and Robertson. We first give a negative answer to their uniform $B + mB$ partition question: the fixed 3-colouring $\chi(n) = \lfloor \log_2 n \rfloor \pmod 3$ has the property that, for every $q \in \mathbb{N}$, some m coprime to q admits no infinite $B \subseteq \mathbb{N}$ for which $B + mB$ is monochromatic.

Our fibre results give a negative answer to their Cartesian-product question: for every $\varepsilon > 0$ there is a symmetric set $A \subseteq \mathbb{N}^2$ whose horizontal and vertical fibres all have lower density at least $1 - \varepsilon$, but no translate of the ordered upper triangle $\{(b_i, b_j) : i < j\}$ of an infinite set is contained in A. The same shifted-difference construction gives a negative answer, already in $G = (\mathbb{Z}, +)$, to the amenable corner-sumset question, again with all horizontal and vertical fibres of symmetric lower density arbitrarily close to one.

We also prove sharp density-one boundary results. For the Cartesian question, a density-one good-rows criterion gives a positive theorem, showing that the uniform fibre-density threshold is exactly one. For the amenable corner question over $\mathbb{Z}$, vertical density one alone is insufficient, but an anchor-row density-one criterion gives a positive theorem; in particular, if all horizontal and vertical fibres have density one, then the desired infinite corner-sumset exists.

Finally we record a separate finite-fibre obstruction to the symmetric product-rectangle problem $BC \cup CB \subseteq A$ in amenable groups. In generalized dihedral groups $\mathbb{Z}^d \rtimes C_2$, order cones in the reflection layer give positive-density sets containing no $BC \cup CB$ with B, C infinite. We give a parity reduction in D_∞, a conjugation-invariant positive contrast, and a reduction of the sharp density threshold in D_∞ to a one-dimensional sum-difference threshold.

Contents

Date: June 9, 2026.

1. Introduction

A theorem of Moreira, Richter and Robertson [16, Theorem 1.3] asserts that every subset of a countably infinite amenable group with positive upper density along a Følner sequence contains a one-sided product set BC with B, C infinite. In the integers this is the additive assertion that every positive-density set contains a sumset $B+C$ with $B, C \subseteq \mathbb{N}$ infinite. Kra, Moreira, Richter and Robertson subsequently formulated many refinements and analogues of this phenomenon. This paper gives a negative answer to the uniform partition refinement [14, Question 3.20]; gives negative answers, together with density-one boundary results, for [14, Questions 5.10 and 5.23]; and gives a negative answer in generalized dihedral groups to the symmetric product-rectangle problem [12, Question 8.7].

The questions considered here lie at the interface of Ramsey theory, additive combinatorics, and ergodic theory. Hindman's finite sums theorem [8] is the partition-regular benchmark, while Erdős's density sumset problem [3] motivated early partial results of Nathanson [17] and was later resolved by Moreira, Richter and Robertson [16]; see also Host's short ergodic proof [9]. Kra, Moreira, Richter and Robertson developed several stronger positive results, including restricted-sumset and density finite-sums theorems [13, 15], and their survey [14] organizes many of the refinements addressed here. Recent related work includes Ackelsberg's Straus-type counterexamples [1], results on asymmetric and unrestricted sumsets [10, 11], density Hindman-type and infinite linear configurations [4, 6], growth and recurrence refinements [5, 7], and product-set results in classes of amenable groups [2]. The present paper is complementary to this positive theory: it isolates high-density obstructions to several proposed product-type refinements, while keeping the positive density-one boundary explicit.

The first result concerns the partition question. For $m \in \mathbb{N}$ and $B \subseteq \mathbb{N}$, write

$$B + mB = \{b_1 + mb_2 : b_1, b_2 \in B\}.$$

[14, Question 3.20] asks whether every finite colouring of $\mathbb{N}$ admits a number $q \in \mathbb{N}$ such that, for every m coprime to q, some infinite $B \subseteq \mathbb{N}$ has $B + mB$ monochromatic. We show that the answer is negative.

Theorem 1.1. *Define $\chi : \mathbb{N} \to \mathbb{Z}/3\mathbb{Z}$ by*

$$\chi(n) = \lfloor \log_2 n \rfloor \pmod 3.$$

For every $q \in \mathbb{N}$ there exists $m \in \mathbb{N}$ with $(m, q) = 1$ such that no infinite $B \subseteq \mathbb{N}$ has $B + mB$ monochromatic with respect to χ.

The proof is given in Section 2. In fact, every m in the logarithmic blocks

$$\bigcup_{r \geq 0} (2^{3r+1}, 2^{3r+2}) \cap \mathbb{N}$$

is bad for this fixed colouring. This settles [14, Question 3.20], but it does not settle the weaker [14, Question 3.19] asking only whether some value of m works for every finite colouring.

The next direction is the Cartesian-product question, namely [14, Question 5.10]. For $A \subseteq \mathbb{N}^2$, write

$$A_x = \{y \in \mathbb{N} : (x, y) \in A\}$$

for the vertical fibre over x. The survey asks whether a uniform positive lower-density assumption on the fibres A_x forces a translate of the ordered upper triangle

$$\{(b_i, b_j) : i < j\}$$

of an infinite set $B = \{b_1 < b_2 < \cdots\} \subseteq \mathbb{N}$.

Our fibre obstruction shows that the answer is negative in a strong form.

Theorem 1.2. *For every $\varepsilon > 0$ there exists a symmetric set $A \subseteq \mathbb{N}^2$ such that every vertical and every horizontal fibre of A has natural lower density at least $1-\varepsilon$, but there are no infinite $B = \{b_1 < b_2 < \cdots\} \subseteq \mathbb{N}$ and no shift $(u, v) \in \mathbb{Z}^2$ such that*

$$(b_i + u, b_j + v) \in A \qquad (i < j).$$

For typographical reasons the displayed result is restated and proved below as Theorem 5.1. In particular, [14, Question 5.10] has a negative answer even under diagonal symmetry and with all horizontal and vertical fibres of lower density arbitrarily close to one. The construction is diagonal: we build a high-density set $T \subseteq \mathbb{Z}$ such that every shift $T - h$ avoids the complete ordered difference set of every injective sequence, and then put

$$A = \{(x, y) : y - x \in T\}.$$

The obstruction is a shifted version of the Straus deletion argument: for each possible shift h we choose a prime p_h and remove all non-zero p_h-multiples after translating by h.

The same construction applies to the amenable corner-sumset question [14, Question 5.23]. In additive notation for $G = (\mathbb{Z}, +)$, that question asks whether a large $A \subseteq \mathbb{Z}^2$ must contain

$$(t, s), \qquad (b_i + t, s), \qquad (b_i + t, b_j + s) \quad (i < j)$$

for some $(t, s) \in \mathbb{Z}^2$ and some injective sequence $b : \mathbb{N} \to \mathbb{Z}$.

Theorem 1.3. *For every $\varepsilon > 0$ there exists a symmetric set $A \subseteq \mathbb{Z}^2$ such that every vertical and every horizontal fibre has symmetric lower density at least $1-\varepsilon$, but no such corner-sumset configuration exists.*

This is Theorem 5.3. Thus [14, Question 5.23] is false already for $G = (\mathbb{Z}, +)$. It is worth contrasting this with an even simpler obstruction: one can have vertical fibres of density one and two-dimensional density one, while every horizontal fibre is finite. This alone already refutes the corner question as stated, but it does not address any strengthened formulation with horizontal-fibre hypotheses. The shifted-difference construction does address such strengthened formulations below the density-one threshold.

For the fibre questions, the next theme is sharpness. The negative examples above have fibre densities arbitrarily close to one. We prove that the threshold one itself is positive. For the Cartesian question, a density-one good-rows condition suffices. A simple special case is the following.

Theorem 1.4. *Let $A \subseteq \mathbb{N}^2$. If every vertical fibre A_x has lower density one along $[N] = \{1, \ldots, N\}$, then there is an infinite set $B = \{b_1 < b_2 < \cdots\} \subseteq \mathbb{N}$ such that*

$$(b_i, b_j) \in A \qquad (i < j).$$

The more flexible statement, Theorem 6.4, allows a density-one set of good rows at each error scale. Together with Theorem 5.1, it shows that the uniform fibre-density threshold for the Cartesian question is exactly one.

For the amenable corner-sumset question on $\mathbb{Z}$, vertical density one alone is not enough, because the points $(b_i + t, s)$ all lie on one horizontal line. However, an anchor-row density-one hypothesis suffices. In particular, if every horizontal and every vertical fibre has density one, then the desired configuration exists; see Theorem 7.1 and Corollary 7.2.

The final part of the paper records a related obstruction to the symmetric product-rectangle question [12, Question 8.7],

$$BC \cup CB \subseteq A$$

in amenable groups. This question is stronger than the one-sided product-set theorem. In generalized dihedral groups $G_d = \mathbb{Z}^d \rtimes C_2$, an order cone in the reflection layer gives a positive-density counterexample, so [12, Question 8.7] has a negative answer. We also show that conjugation-invariance of A restores the two-sided conclusion, and we reduce the sharp threshold problem in D_∞ to a one-dimensional sum-difference problem.

Remark 1.5 (Scope and priority)**.** The paper deliberately does not use as new results several statements that are now covered by recent work of Ackelsberg, Hernandez, Hetzel, Hernandez–Kousek–Radic, and Hernandez–Hetzel. In particular, the polynomial ordered-sum and ordered product-sum questions are not claimed here as new negative answers. The self-contained contributions here are the logarithmic partition obstruction, the fibre and product-rectangle obstructions, and the density-one boundary criteria proved below.

2. Proof of Theorem 2.2

Given $m \in \mathbb{N}$ and $B \subseteq \mathbb{N}$, write

$$B + mB = \{b_1 + mb_2 : b_1, b_2 \in B\}.$$

Kra, Moreira, Richter and Robertson ask whether every finite colouring of $\mathbb{N}$ has a cell containing $B + mB$ for some $m \in \mathbb{N}$ and some infinite $B \subseteq \mathbb{N}$ [14, Question 3.19]. They then ask whether the possible obstructions in m can be confined to finitely many prime factors.

Question 2.1 (Kra–Moreira–Richter–Robertson, Question 3.20)**.** *For every finite colouring of $\mathbb{N}$, does there exist $q \in \mathbb{N}$ such that for all $m \in \mathbb{N}$ with $(m, q) = 1$ there is an infinite set $B \subseteq \mathbb{N}$ for which $B + mB$ is monochromatic?*

We give a negative answer to Question 2.1. The counterexample is a single logarithmic block colouring. It is closely related in spirit to the colouring used in [14, Example 3.9], where for each fixed m a colouring depending on m is constructed to avoid monochromatic configurations of the form $B + mB$. The point here is that one fixed colouring suffices to produce a bad value of m in every coprimality class.

Theorem 2.2 (Logarithmic colouring obstruction)**.** *Define $\chi : \mathbb{N} \to \mathbb{Z}/3\mathbb{Z}$ by*

$$\chi(n) = \lfloor \log_2 n \rfloor \pmod 3.$$

For every $q \in \mathbb{N}$ there exists $m \in \mathbb{N}$ with $(m, q) = 1$ such that no infinite set $B \subseteq \mathbb{N}$ has $B + mB$ monochromatic with respect to χ. Consequently, Question 2.1 has a negative answer.

We prove a little more. Let

$$\mathcal{M} = \bigcup_{r \geq 0} (2^{3r+1}, 2^{3r+2}) \cap \mathbb{N}.$$

For every $m \in \mathcal{M}$ and every infinite $B \subseteq \mathbb{N}$, the set $B + mB$ contains two elements with different χ-colours.

The mechanism is simple. Fix $a \in B$ and let $b \in B$ tend to infinity. Then

$$b + ma \sim b, \qquad a + mb \sim mb,$$

so the two elements $b + ma$ and $a + mb$ of $B + mB$ are separated on the logarithmic scale by approximately $\log_2 m$. When $\log_2 m$ lies strictly between $3r + 1$ and $3r + 2$, this separation changes the value of $\lfloor \log_2(\cdot) \rfloor$ by either $3r + 1$ or $3r + 2$, and hence changes the colour modulo 3.

Lemma 2.3. *Let $r \in \mathbb{Z}$ and let $\Delta \in (3r + 1, 3r + 2)$. Then, for every real number X,*

$$\lfloor X + \Delta \rfloor - \lfloor X \rfloor \in \{3r + 1, 3r + 2\}.$$

In particular,

$$\lfloor X + \Delta \rfloor \not\equiv \lfloor X \rfloor \pmod 3.$$

Proof. Write $X = n + u$ with $n \in \mathbb{Z}$ and $u \in [0, 1)$. Also write $\Delta = 3r + 1 + s$ with $s \in (0, 1)$. Then

$$\lfloor X + \Delta \rfloor - \lfloor X \rfloor = \lfloor u + 3r + 1 + s \rfloor = 3r + 1 + \lfloor u + s \rfloor.$$

Since $u + s \in (0, 2)$, the last floor is either 0 or 1. This proves the claim. □

Lemma 2.4. *For every $q \in \mathbb{N}$ there exists $m \in \mathcal{M}$ with $(m, q) = 1$.*

Proof. The interval $(2^{3r+1}, 2^{3r+2})$ has length 2^{3r+1}, which tends to infinity with r. Choose r so large that this length is greater than $q + 1$; then the open interval contains q consecutive integers. Among any q consecutive integers there is a complete system of residues modulo q, and at least one residue class is coprime to q. Hence the interval contains some integer m with $(m, q) = 1$. □

Proposition 2.5. *Let*

$$\chi(n) = \lfloor \log_2 n \rfloor \pmod 3.$$

If $m \in \mathcal{M}$, then there is no infinite set $B \subseteq \mathbb{N}$ such that $B + mB$ is monochromatic with respect to χ.

Proof. Let $m \in \mathcal{M}$. Then for some $r \geq 0$,

$$2^{3r+1} < m < 2^{3r+2}.$$

Equivalently,

$$\alpha = \log_2 m \in (3r + 1, 3r + 2).$$

Assume, toward a contradiction, that $B \subseteq \mathbb{N}$ is infinite and that $B + mB$ is monochromatic.

Fix $a \in B$. Since B is infinite, it is unbounded. Choose $b \in B$ with $b > a$ and large enough that

$$\left| \log_2 \left(\frac{a + mb}{b + ma} \right) - \log_2 m \right| < \eta,$$

where

$$\eta = \frac{1}{2} \min\{\alpha - (3r + 1), (3r + 2) - \alpha\} > 0.$$

This is possible because

$$\frac{a + mb}{b + ma} \longrightarrow m \qquad \text{as } b \to \infty.$$

Put

$$x = b + ma, \qquad y = a + mb.$$

Both x and y belong to $B + mB$. Moreover,

$$\Delta = \log_2 y - \log_2 x = \log_2 \left(\frac{a + mb}{b + ma} \right)$$

satisfies

$$\Delta \in (3r + 1, 3r + 2).$$

Applying Lemma 2.3 with $X = \log_2 x$, we obtain

$$\lfloor \log_2 y \rfloor \not\equiv \lfloor \log_2 x \rfloor \pmod 3.$$

Thus

$$\chi(y) \neq \chi(x),$$

contradicting the assumption that $B + mB$ is monochromatic. Therefore no such infinite set B exists. □

Proof of Theorem 2.2. Let $q \in \mathbb{N}$ be arbitrary. By Lemma 2.4, choose $m \in \mathcal{M}$ with $(m, q) = 1$. By Proposition 2.5, no infinite $B \subseteq \mathbb{N}$ has $B + mB$ monochromatic. Hence the colouring χ has the property that, for every q, some m coprime to q is bad. This is precisely the negation of Question 2.1. □

Remark 2.6 (Logarithmically thick exceptional parameters)**.** The bad set of parameters

$$\mathcal{M} = \bigcup_{r \geq 0} (2^{3r+1}, 2^{3r+2}) \cap \mathbb{N}$$

is not sparse on the logarithmic scale. In each interval $(2^{3r}, 2^{3r+3})$, it occupies one of the three dyadic decades. The negative answer to Question 2.1 therefore does not come from a thin exceptional sequence of parameters m.

Remark 2.7 (Why this does not settle Question 3.19)**.** The colouring in Theorem 2.2 does not rule out the possibility that there exists some m and some infinite B for which $B + mB$ is monochromatic. For example, the preceding argument gives no obstruction when $m = 2^{3r}$, since multiplication by m shifts $\lfloor \log_2(\cdot) \rfloor$ by approximately a multiple of 3. Thus the proof gives a negative answer to the uniform coprimality refinement, Question 2.1, but not to the weaker question asking only for the existence of one good m.

Remark 2.8 (Other bases)**.** The base 2 is inessential. If $a \geq 2$ is an integer and one colours by

$$\chi_a(n) = \lfloor \log_a n \rfloor \pmod 3,$$

then every m with

$$a^{3r+1} < m < a^{3r+2}$$

is bad for the same proof. Since such intervals again contain integers coprime to any prescribed q when r is large, this gives many analogous counterexamples.

3. Definitions and notation

We write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$. For $E \subseteq \mathbb{N}$ define the lower asymptotic density

$$\underline{d}(E) = \liminf_{N \to \infty} \frac{|E \cap [1, N]|}{N}.$$

For $E \subseteq \mathbb{Z}$ define the symmetric lower density

$$\underline{d}_{\mathrm{sym}}(E) = \liminf_{N \to \infty} \frac{|E \cap [-N, N]|}{2N + 1}.$$

When the corresponding limit exists we write $d(E)$ or $d_{\mathrm{sym}}(E)$.

All Følner sequences in this paper consist of finite non-empty sets with cardinalities tending to infinity. If $\Phi = (\Phi_N)$ is a Følner sequence in a countable amenable group or semigroup and E is a subset, write

$$\overline{d}_\Phi(E) = \limsup_{N\to\infty} \frac{|E \cap \Phi_N|}{|\Phi_N|}, \qquad \underline{d}_\Phi(E) = \liminf_{N\to\infty} \frac{|E \cap \Phi_N|}{|\Phi_N|}.$$

For an increasing infinite set $B = \{b_1 < b_2 < \cdots\} \subseteq \mathbb{N}$ define its ordered positive difference set by

$$\Delta(B) = \{b_j - b_i : i < j\}.$$

For an injective sequence $b : \mathbb{N} \to \mathbb{Z}$ define

$$\Delta(b) = \{b(j) - b(i) : i < j\}.$$

These differences need not be positive.

For $A \subseteq \mathbb{N}^2$ or $A \subseteq \mathbb{Z}^2$, the vertical and horizontal fibres are

$$A_x = \{y : (x, y) \in A\}, \qquad H_y = \{x : (x, y) \in A\}.$$

For a subset E of an additive group and an element t, we use

$$E - t = \{x : x + t \in E\}.$$

For a finite set F we write

$$E - F = \bigcup_{f\in F} (E - f).$$

When the ambient set is $\mathbb{N}$, the variable x is required to lie in $\mathbb{N}$.

We shall also use a one-sided version of piecewise syndeticity for subsets of $\mathbb{Z}$.

Definition 3.1. A set $D \subseteq \mathbb{Z}$ is *right thick* if for every $L \in \mathbb{N}$ and every $M \in \mathbb{Z}$ there is $n \geq M$ such that

$$[n, n + L] \subseteq D.$$

A set $T \subseteq \mathbb{Z}$ is *right piecewise syndetic* if there is a finite set $F \subseteq \mathbb{Z}$ such that

$$T - F := \bigcup_{f\in F} (T - f)$$

is right thick.

Right piecewise syndeticity is the notion relevant to positive ordered differences in $\mathbb{N}$. The usual group-theoretic notion of thickness in $\mathbb{Z}$ allows intervals drifting to $-\infty$, which is not sufficient for the positive-difference configurations considered in the Cartesian question over $\mathbb{N}$.

4. Proof of Theorem 4.1

This section contains the central deletion construction.

Theorem 4.1 (High-density shifted-difference obstruction)**.** *For every $\varepsilon > 0$ there exists a symmetric set $T = -T \subseteq \mathbb{Z}$ with the following properties.*

(i) For every fixed $a \in \mathbb{Z}$,

$$\underline{d}\big((a + T) \cap \mathbb{N}\big) > 1 - \varepsilon, \qquad \underline{d}_{\mathrm{sym}}(a + T) > 1 - \varepsilon.$$

(ii) For every $h \in \mathbb{Z}$, there is a prime p_h such that

$$(T - h) \cap p_h\mathbb{Z} \subseteq \{0\}.$$

(iii) Consequently, for every $h \in \mathbb{Z}$ and every injective sequence $b : \mathbb{N} \to \mathbb{Z}$,

$$\Delta(b) \not\subseteq T - h.$$

In particular, for every infinite $B \subseteq \mathbb{N}$,

$$\Delta(B) \not\subseteq T - h.$$

Proof. Choose primes $(p_h)_{h\in\mathbb{Z}}$ such that

$$p_{-h} = p_h, \qquad p_h > 10(1+|h|)^2, \qquad \sum_{h\in\mathbb{Z}} \frac{1}{p_h} < \varepsilon.$$

This is possible because primes are unbounded; choose first p_0 and then choose the primes for pairs $\{h, -h\}$ sufficiently large.

Define

$$D = \bigcup_{h\in\mathbb{Z}} \{h + kp_h : k \in \mathbb{Z},\ k \neq 0\}, \qquad T = \mathbb{Z} \setminus D.$$

Because $p_{-h} = p_h$, the set D is symmetric: if $x = h + kp_h$ with $k \neq 0$, then

$$-x = (-h) + (-k)p_{-h} \in D.$$

Thus $T = -T$.

We estimate density in translated one-sided intervals; the symmetric estimate is identical. Fix $a \in \mathbb{Z}$ and set

$$I_N = [1-a, N-a] \cap \mathbb{Z}.$$

Let

$$H_N = \{h \in \mathbb{Z} : \exists k \in \mathbb{Z},\ k \neq 0 \text{ with } h + kp_h \in I_N\}.$$

If $h \in H_N$, then for such a non-zero k,

$$p_h - |h| \leq |h + kp_h| \leq N + |a|.$$

Since $p_h > 10(1+|h|)^2$, this can happen only for $|h| = O_a(\sqrt{N})$, apart from changing the implicit constant. Hence $|H_N| = O_a(\sqrt{N})$.

For fixed h, the progression $h + p_h\mathbb{Z}$ meets I_N in at most

$$\frac{N}{p_h} + O(1)$$

points. Therefore

$$|D \cap I_N| \leq \sum_{h\in H_N} \left(\frac{N}{p_h} + O(1)\right) \leq N \sum_{h\in\mathbb{Z}} \frac{1}{p_h} + O_a(\sqrt{N}).$$

Dividing by N and taking the limsup gives

$$\limsup_{N\to\infty} \frac{|D \cap I_N|}{N} \leq \sum_h \frac{1}{p_h} < \varepsilon.$$

Thus

$$\underline{d}((a+T) \cap \mathbb{N}) > 1 - \varepsilon.$$

The symmetric estimate for $[-N-a, N-a]$ gives $\underline{d}_{\mathrm{sym}}(a+T) > 1-\varepsilon$.

Now fix $h \in \mathbb{Z}$. If $x = kp_h$ with $k \neq 0$, then

$$x + h = h + kp_h \in D,$$

so $x + h \notin T$. Hence no non-zero element of $p_h\mathbb{Z}$ belongs to $T - h$, and

$$(T-h) \cap p_h\mathbb{Z} \subseteq \{0\}.$$

Finally let $b : \mathbb{N} \to \mathbb{Z}$ be injective. By the infinite pigeonhole principle, infinitely many values of $b(i)$ lie in a single residue class modulo p_h. Choose $i < j$ from that infinite subsequence. Then $b(j) - b(i)$ is a non-zero multiple of p_h, so it is not in $T - h$. Hence $\Delta(b) \not\subseteq T - h$. $\square$

Remark 4.2. The exclusion of the term $k = 0$ in the deleted set $\{h + kp_h : k \neq 0\}$ is essential. If one deleted the whole residue class $h \pmod{p_h}$ for every $h \in \mathbb{Z}$, then every integer would be deleted by its own shift.

5. Proofs of Theorems 5.1 and 5.3

5.1. The Cartesian-product question.

Theorem 5.1 (Negative answer to the Cartesian question)**.** *For every $\varepsilon > 0$ there exists a symmetric set $A \subseteq \mathbb{N}^2$ such that every vertical and horizontal fibre has lower natural density greater than $1 - \varepsilon$, but there are no infinite $B = \{b_1 < b_2 < \cdots\} \subseteq \mathbb{N}$ and no shift $(u, v) \in \mathbb{Z}^2$ such that*

$$(b_i + u, b_j + v) \in A \qquad (i < j).$$

Proof. Let $T \subseteq \mathbb{Z}$ be obtained from Theorem 4.1. Define

$$A = \{(x, y) \in \mathbb{N}^2 : y - x \in T\}.$$

Since $T = -T$, the set A is symmetric under interchanging the coordinates. For fixed $x \in \mathbb{N}$,

$$A_x = (x + T) \cap \mathbb{N},$$

which has lower density greater than $1 - \varepsilon$ by Theorem 4.1. Horizontal fibres have the same lower bound by symmetry.

Suppose, toward a contradiction, that $B = \{b_1 < b_2 < \cdots\}$ and $(u, v) \in \mathbb{Z}^2$ satisfy

$$(b_i + u, b_j + v) \in A \qquad (i < j).$$

Then

$$(b_j + v) - (b_i + u) = b_j - b_i + (v - u) \in T$$

for all $i < j$. Therefore

$$\Delta(B) \subseteq T - (v - u),$$

contradicting Theorem 4.1. $\square$

Corollary 5.2. *The Cartesian fibre-density question has a negative answer even under diagonal symmetry and even if all vertical and horizontal fibres are required to have lower density at least $1 - \varepsilon$ for an arbitrarily prescribed $\varepsilon > 0$.*

5.2. The amenable corner-sumset question over $\mathbb{Z}$.

Theorem 5.3 (High-density negative answer to the amenable corner question)**.** *For every $\varepsilon > 0$ there exists a symmetric set $A \subseteq \mathbb{Z}^2$ such that every vertical and horizontal fibre has symmetric lower density greater than $1-\varepsilon$, but there are no $(t, s) \in \mathbb{Z}^2$ and no injective sequence $b : \mathbb{N} \to \mathbb{Z}$ satisfying*

$$(t, s), \quad (b(i) + t, s), \quad (b(i) + t, b(j) + s) \in A \qquad (i < j).$$

Proof. Let T be the set from Theorem 4.1 and define

$$A = \{(x, y) \in \mathbb{Z}^2 : y - x \in T\}.$$

Every horizontal and vertical fibre is a translate of T, hence has symmetric lower density greater than $1 - \varepsilon$.

If the displayed configuration existed, then using only the third family of points gives

$$(b(j)+s)-(b(i)+t)=b(j)-b(i)+(s-t)\in T \qquad (i<j).$$

Thus

$$\Delta(b)\subseteq T-(s-t),$$

contradicting Theorem 4.1. $\square$

5.3. **A finite-horizontal-fibre obstruction.** The previous theorem rules out even symmetric high-fibre variants below the density-one threshold. The original amenable corner question is also false for a still simpler reason: the hypothesis controls vertical fibres but the conclusion forces infinitely many points into one horizontal fibre.

Proposition 5.4 (A full-density finite-horizontal-fibre example). *Let $G=(\mathbb{Z},+)$, let $\Phi_N=[-N,N]$, and define*

$$A=\{(x,y)\in\mathbb{Z}^2:|x|\le(|y|+1)^2\}.$$

Then every vertical fibre has density one along Φ_N, and

$$\lim_{N\to\infty}\frac{|A\cap([-N,N]^2)|}{(2N+1)^2}=1.$$

Nevertheless there are no $(t,s)\in\mathbb{Z}^2$ and no injective $b:\mathbb{N}\to\mathbb{Z}$ satisfying

$$(t,s),\quad (b(i)+t,s),\quad (b(i)+t,b(j)+s)\in A \qquad (i<j).$$

Proof. For fixed $g\in\mathbb{Z}$ the vertical fibre is

$$A_g=\{h\in\mathbb{Z}:|g|\le(|h|+1)^2\}.$$

Its complement is finite, so it has density one.

For the two-dimensional density, if $(x,y)\in[-N,N]^2\setminus A$, then

$$|x|>(|y|+1)^2$$

and $|x|\le N$, so $|y|<\sqrt{N}$. Thus the complement in the square has size $O(N^{3/2})=o(N^2)$.

If the corner-sumset configuration existed, then from $(b(i)+t,s)\in A$ we would get

$$|b(i)+t|\le(|s|+1)^2$$

for every i. Hence all $b(i)$ lie in a finite interval of $\mathbb{Z}$, contradicting injectivity. $\square$

Remark 5.5. This finite-horizontal-fibre example does not refute the Cartesian question of Theorem 5.1; the points $(b(i)+t,s)$ on a fixed horizontal line are part of the corner-sumset configuration, not part of the shifted ordered upper triangle in the Cartesian question.

6. Proofs of Theorems 6.2 and 6.4

The high-density counterexample shows that any uniform density threshold below one is insufficient. We now prove positive results at the density-one boundary.

Proposition 6.1 (Self finite-intersection formulation). *Let $A\subseteq\mathbb{N}^2$. There exist $(u,v)\in\mathbb{N}^2$ and an infinite $B=\{b_1<b_2<\cdots\}\subseteq\mathbb{N}$ such that*

$$(b_i+u,b_j+v)\in A \qquad (i<j)$$

if and only if there exist $(u,v)\in\mathbb{N}^2$ and an infinite $X\subseteq\mathbb{N}$ such that for every finite $F\subseteq X$,

$$X\cap\bigcap_{x\in F}(A_{x+u}-v)$$

is infinite.

Proof. If B is a desired sequence, take $X = B$. For any finite $F \subseteq B$, all sufficiently late elements of B lie in $A_{x+u} - v$ for every $x \in F$.

Conversely, suppose X, u, v have the stated property. Choose $b_1 \in X$. Having chosen $b_1 < \cdots < b_n$ in X, choose

$$b_{n+1} > b_n$$

inside

$$X \cap \bigcap_{i=1}^{n} (A_{b_i+u} - v),$$

which is infinite. Then $(b_i + u, b_{n+1} + v) \in A$ for all $i \leq n$. The recursion gives the desired sequence. □

Theorem 6.2 (Finite-shift intersection criterion)**.** *Let $A \subseteq \mathbb{N}^2$. Suppose there is a finite set $F \subseteq \mathbb{N}_0$ such that for every finite $X \subseteq \mathbb{N}$,*

$$\bigcap_{x \in X} (A_x - F)$$

is infinite, where

$$A_x - F = \{y \in \mathbb{N} : \exists f \in F \text{ with } y + f \in A_x\}.$$

Then there are an infinite $B = \{b_1 < b_2 < \cdots\} \subseteq \mathbb{N}$ and a single $f_0 \in F$ such that

$$(b_i, b_j + f_0) \in A \qquad (i < j).$$

Equivalently,

$$\{(b_i, b_j) : i < j\} \subseteq A - (0, f_0).$$

Proof. Choose any $b_1 \in \mathbb{N}$. Construct $b_1 < b_2 < \cdots$ recursively. Having chosen $b_1 < \cdots < b_n$, choose

$$b_{n+1} > b_n$$

in

$$\bigcap_{i=1}^{n} (A_{b_i} - F),$$

which is infinite. For every $i < n + 1$, choose $f_{i,n+1} \in F$ such that

$$b_{n+1} + f_{i,n+1} \in A_{b_i}.$$

Thus each pair $i < j$ receives a colour $f_{i,j} \in F$. By the infinite Ramsey theorem [18], pass to an infinite subsequence on which this colour is constant, say f_0. For that subsequence,

$$b_j + f_0 \in A_{b_i} \qquad (i < j),$$

as required. □

Corollary 6.3 (Uniformly syndetic vertical fibres)**.** *If there is a finite $F \subseteq \mathbb{N}_0$ such that*

$$A_x - F = \mathbb{N} \qquad \text{for every } x \in \mathbb{N},$$

then the Cartesian conclusion holds.

Theorem 6.4 (Density-one good rows)**.** *Let Φ be a Følner sequence on $\mathbb{N}$ and let $A \subseteq \mathbb{N}^2$. Let $(\varepsilon_i)_{i \geq 1}$ be positive numbers with*

$$\sum_{i=1}^{\infty} \varepsilon_i < 1.$$

For each i, define

$$H_i = \{n \in \mathbb{N} : \underline{d}_\Phi(A_n) > 1 - \varepsilon_i\}.$$

If

$$\underline{d}_\Phi(H_i) = 1 \qquad \textit{for every } i,$$

then there exists an infinite $B = \{b_1 < b_2 < \cdots\} \subseteq \mathbb{N}$ *such that*

$$(b_i, b_j) \in A \qquad (i < j).$$

Proof. Choose $b_1 \in H_1$. Suppose $b_1 < \cdots < b_k$ have been chosen with $b_i \in H_i$. Put

$$C_k = \bigcap_{i=1}^{k} A_{b_i}.$$

By the union bound along Φ,

$$\underline{d}_\Phi(C_k) \geq 1 - \sum_{i=1}^{k} \varepsilon_i > 0.$$

Since $\underline{d}_\Phi(H_{k+1}) = 1$, the set $C_k \cap H_{k+1}$ has positive lower Φ-density, hence is infinite. Choose

$$b_{k+1} > b_k$$

inside this intersection. Then $b_{k+1} \in A_{b_i}$ for every $i \leq k$, so $(b_i, b_{k+1}) \in A$ for all $i \leq k$. The recursion completes the proof. □

Corollary 6.5 (Sharp uniform fibre-density threshold)**.** *In the sense of a uniform lower bound on fibre densities, the Cartesian question has threshold exactly one. If all rows satisfy a density-one hypothesis as in Theorem 6.4, then the conclusion holds. For every* $\delta < 1$, *Theorem 5.1 gives a symmetric counterexample with all horizontal and vertical fibres of lower density at least* δ.

6.1. **Diagonal sets and right piecewise syndeticity.** For $T \subseteq \mathbb{Z}$ define

$$A_T^+ = \{(x, y) \in \mathbb{N}^2 : y - x \in T\}.$$

Proposition 6.6 (Diagonal reduction for the Cartesian question)**.** *For* A_T^+, *the Cartesian conclusion holds if and only if there exist* $h \in \mathbb{Z}$ *and an infinite* $B = \{b_1 < b_2 < \cdots\} \subseteq \mathbb{N}$ *such that*

$$\Delta(B) \subseteq T - h.$$

Proof. If $(b_i + u, b_j + v) \in A_T^+$ for all $i < j$, then

$$b_j - b_i + (v - u) \in T,$$

so $\Delta(B) \subseteq T - (v - u)$.

Conversely, suppose $\Delta(B) \subseteq T - h$. Choose $u, v \in \mathbb{N}$ with $v - u = h$. Then

$$(b_j + v) - (b_i + u) = b_j - b_i + h \in T,$$

so $(b_i + u, b_j + v) \in A_T^+$ for all $i < j$. □

Theorem 6.7 (Right piecewise syndetic diagonal sets are positive)**.** *If* $T \subseteq \mathbb{Z}$ *is right piecewise syndetic, then* A_T^+ *satisfies the Cartesian conclusion.*

Proof. Choose a finite set $F \subseteq \mathbb{Z}$ such that

$$D = T - F$$

is right thick. We first construct an infinite increasing $B_0 = \{c_1 < c_2 < \cdots\} \subseteq \mathbb{N}$ with

$$\Delta(B_0) \subseteq D.$$

Start with any $c_1 \in \mathbb{N}$. Having chosen $c_1 < \cdots < c_n$, the finite intersection

$$\bigcap_{i=1}^{n}(D + c_i)$$

is right thick, hence contains arbitrarily large integers. Indeed, this uses that all the sets are translates of the same right thick set. If $c_{\min} = \min_i c_i$, $c_{\max} = \max_i c_i$, and $C = c_{\max} - c_{\min}$, then for any L and M we may choose $r \geq M - c_{\max}$ with

$$[r, r + L + C] \subseteq D.$$

Setting $m = r + c_{\max}$ gives

$$[m, m + L] \subseteq \bigcap_{i=1}^{n}(D + c_i).$$

Choose $c_{n+1} > c_n$ in that intersection. Then $c_{n+1} - c_i \in D$ for every $i \leq n$.

For each difference $d = c_j - c_i \in \Delta(B_0) \subseteq D = T - F$, choose $f(d) \in F$ such that $d + f(d) \in T$. Colour pairs $i < j$ by $f(c_j - c_i)$. Infinite Ramsey [18] gives an infinite subsequence $B \subseteq B_0$ and a fixed $h \in F$ such that

$$b_j - b_i + h \in T \qquad (i < j,\ b_i, b_j \in B).$$

Thus $\Delta(B) \subseteq T - h$, and Proposition 6.6 applies. ☐

Proposition 6.8 (The constructed obstruction is not right piecewise syndetic)**.** *Let T be constructed as in Theorem 4.1. Then T is not right piecewise syndetic. In fact, for every non-empty finite $F \subseteq \mathbb{Z}$, the set $T - F$ misses all sufficiently large multiples of a fixed integer.*

Proof. The empty set F plainly does not witness right piecewise syndeticity, since then $T - F = \varnothing$. Fix a non-empty finite $F \subseteq \mathbb{Z}$. For each $f \in F$, Theorem 4.1 gives

$$(T - f) \cap p_f\mathbb{Z} \subseteq \{0\}.$$

Let

$$P_F = \prod_{f \in F} p_f.$$

If n is a non-zero multiple of P_F, then n is a non-zero multiple of p_f for every $f \in F$, hence $n \notin T - f$ for every $f \in F$. Therefore

$$n \notin T - F.$$

Thus $T - F$ misses all non-zero multiples of P_F. Every interval of length P_F that is far enough to the right contains a positive multiple of P_F, so $T - F$ cannot be right thick. Since F was arbitrary, T is not right piecewise syndetic. ☐

7. Proof of Theorem 7.1

7.1. **Vertical density one alone is not enough.** The finite-horizontal-fibre example already shows that vertical density one alone cannot force the corner-sumset configuration. A simpler variant is

$$A = \{(x, y) \in \mathbb{Z}^2 : |x| \leq |y|\}.$$

Each vertical fibre is cofinite. But if $(b(i) + t, s) \in A$ for infinitely many distinct $b(i)$, then $|b(i) + t| \leq |s|$ for infinitely many i, impossible.

7.2. **An anchor-row criterion.**

Theorem 7.1 (Anchor-row density-one criterion)**.** *Let $A \subseteq \mathbb{Z}^2$. For $s \in \mathbb{Z}$ write*

$$H_s = \{x \in \mathbb{Z} : (x, s) \in A\},$$

and for $x \in \mathbb{Z}$ write

$$V_x = \{y \in \mathbb{Z} : (x, y) \in A\}.$$

Suppose that for some $s \in \mathbb{Z}$ and some positive sequence (ε_i) with $\sum_i \varepsilon_i < 1$, the following hold:

(a) $d_{\mathrm{sym}}(H_s) = 1$;
(b) for each i, the set

$$K_i = \{x \in H_s : \underline{d}_{\mathrm{sym}}(V_x) > 1 - \varepsilon_i\}$$

has symmetric density one.

Then there exist $(t, s) \in \mathbb{Z}^2$ and an injective sequence $b : \mathbb{N} \to \mathbb{Z}$ such that

$$(t, s), \quad (b(i) + t, s), \quad (b(i) + t, b(j) + s) \in A \qquad (i < j).$$

Proof. Choose $t \in H_s$, so $(t, s) \in A$. We recursively construct distinct integers $x_i \in H_s$ and then set

$$b_i = x_i - t.$$

Choose $x_1 \in K_1$. Suppose $x_1, \dots, x_k$ have been chosen with $x_i \in K_i$. We choose x_{k+1} in

$$K_{k+1} \cap \bigcap_{i=1}^{k} \big(V_{x_i} - (s - t)\big).$$

Finite translations preserve symmetric lower density. Since $d_{\mathrm{sym}}(K_{k+1}) = 1$ and $\underline{d}_{\mathrm{sym}}(V_{x_i}) > 1 - \varepsilon_i$, this intersection has lower symmetric density at least

$$1 - \sum_{i=1}^{k} \varepsilon_i > 0.$$

It is therefore infinite, and we may choose x_{k+1} distinct from the previous choices.

Now $(t, s) \in A$. Since each $x_i \in H_s$,

$$(b_i + t, s) = (x_i, s) \in A.$$

If $i < j$, then the choice of x_j gives

$$x_j \in V_{x_i} - (s - t),$$

so

$$(x_i, x_j + s - t) \in A.$$

But $x_i = b_i + t$ and $x_j + s - t = b_j + s$. Hence

$$(b_i + t, b_j + s) \in A.$$

□

Corollary 7.2 (Two-sided density-one boundary)**.** *If every horizontal and every vertical fibre of $A \subseteq \mathbb{Z}^2$ has symmetric density one, then the amenable corner-sumset conclusion holds for $G = (\mathbb{Z}, +)$. Conversely, for every $\delta < 1$, Theorem 5.3 gives a symmetric $A \subseteq \mathbb{Z}^2$ whose horizontal and vertical fibres all have symmetric lower density at least δ, but for which the conclusion fails.*

7.3. **Diagonal sets.** For $T \subseteq \mathbb{Z}$ define

$$A_T = \{(x, y) \in \mathbb{Z}^2 : y - x \in T\}.$$

Proposition 7.3 (Diagonal reduction for the corner question)**.** *For A_T, the corner-sumset conclusion is equivalent to the existence of $h \in \mathbb{Z}$ and an injective sequence $b : \mathbb{N} \to \mathbb{Z}$ such that*

$$h \in T, \qquad h - b(i) \in T \quad (i \in \mathbb{N}), \qquad h + b(j) - b(i) \in T \quad (i < j).$$

Proof. Put $h = s - t$. The three families of points translate as follows:

$$(t, s) \in A_T \iff h = s - t \in T,$$

$$(b(i) + t, s) \in A_T \iff s - (b(i) + t) = h - b(i) \in T,$$

and

$$(b(i) + t, b(j) + s) \in A_T \iff b(j) + s - (b(i) + t) = h + b(j) - b(i) \in T.$$

This is exactly the stated condition. The converse is obtained by choosing any $t \in \mathbb{Z}$ and setting $s = t + h$. □

Proposition 7.4 (Density-one diagonal corner theorem)**.** *If $T \subseteq \mathbb{Z}$ has symmetric density one and is non-empty, then A_T satisfies the corner-sumset conclusion.*

Proof. Choose $h \in T$. Since T has symmetric density one, both

$$h - T = \{b : h - b \in T\}$$

and

$$T - h = \{d : h + d \in T\}$$

have symmetric density one. We recursively construct distinct $b_1, b_2, \ldots$. Having chosen $b_1, \ldots, b_k$, consider

$$(h - T) \cap \bigcap_{i=1}^{k} (b_i + T - h).$$

This is a finite intersection of density-one sets, hence is infinite; choose b_{k+1} from it and outside the previously chosen finite set. Then

$$h - b_{k+1} \in T$$

and

$$h + b_{k+1} - b_i \in T \qquad (i \le k).$$

The recursion gives the conditions in Proposition 7.3. □

8. Proofs of Theorems 8.1, 8.3 and 8.5

We now turn to a related question about products in general amenable groups. The one-sided product-set theorem of Moreira, Richter and Robertson gives $BC \subseteq A$ in every positive-density set in a countably infinite amenable group. A natural strengthening asks whether one can force both orders:

$$BC \cup CB \subseteq A$$

for infinite B, C.

8.1. **Generalized dihedral groups.** Let

$$G_d = \mathbb{Z}^d \rtimes C_2,$$

where the non-trivial element of C_2 acts on $\mathbb{Z}^d$ by inversion. We write elements as (v, ϵ) with $v \in \mathbb{Z}^d$ and $\epsilon \in \{0, 1\}$, and the group law is

$$(v, \epsilon)(w, \delta) = \big(v + (-1)^\epsilon w, \epsilon + \delta \pmod 2\big).$$

For $S \subseteq \mathbb{Z}^d$ and $y \in \mathbb{Z}^d$ define the symmetric fibre

$$I_S(y) = \{x \in \mathbb{Z}^d : y + x \in S \text{ and } y - x \in S\}.$$

Theorem 8.1 (Order-cone obstruction)**.** *Let $S \subseteq \mathbb{Z}^d$ have positive upper density along the boxes $Q_N = [-N, N]^d$, and suppose that $I_S(y)$ is finite for every $y \in \mathbb{Z}^d$. Put*

$$A = S \times \{1\} \subseteq G_d.$$

Then A has positive upper density along the two-sided Følner sequence

$$\Phi_N = Q_N \times \{0, 1\},$$

but there are no infinite $B, C \subseteq G_d$ such that

$$BC \cup CB \subseteq A.$$

Proof. The sequence (Φ_N) is two-sided Følner because left and right multiplication by a fixed element act on each layer by a fixed translate and possibly a reflection of $\mathbb{Z}^d$, while boxes are Følner under translations and reflections.

The density of A along Φ_N is

$$\frac{|A \cap \Phi_N|}{|\Phi_N|} = \frac{|S \cap Q_N|}{2|Q_N|},$$

which has positive limsup by hypothesis.

Suppose $BC \cup CB \subseteq A$ with B, C infinite. Since A lies in the reflection layer $\mathbb{Z}^d \times \{1\}$, for every $b \in B, c \in C$ the parity coordinates of b and c must add to 1 modulo 2. Hence, after no thinning is needed, one of B, C lies in the rotation layer and the other in the reflection layer.

First suppose $B \subseteq \mathbb{Z}^d \times \{0\}$ and $C \subseteq \mathbb{Z}^d \times \{1\}$. Write $b = (x, 0)$ and $c = (y, 1)$. Then

$$bc = (x + y, 1), \qquad cb = (y - x, 1).$$

The condition $bc, cb \in A = S \times \{1\}$ gives

$$y + x \in S, \qquad y - x \in S.$$

Thus $x \in I_S(y)$. Fixing a single $c = (y, 1) \in C$, all x arising from elements of B lie in the finite set $I_S(y)$, contradicting that B is infinite.

The case $B \subseteq \mathbb{Z}^d \times \{1\}$ and $C \subseteq \mathbb{Z}^d \times \{0\}$ is symmetric. Writing $b = (x, 1)$ and $c = (y, 0)$ gives

$$bc = (x - y, 1), \qquad cb = (x + y, 1),$$

so $y \in I_S(x)$. Fixing b forces C to be finite, again a contradiction. $\square$

Corollary 8.2. *For every $d \geq 1$,*

$$A_d = \mathbb{N}_0^d \times \{1\} \subseteq \mathbb{Z}^d \rtimes C_2$$

has density $2^{-(d+1)}$ along $Q_N \times \{0, 1\}$ and contains no $BC \cup CB$ with B, C infinite.

Proof. Take $S = \mathbb{N}_0^d$. If $y \in \mathbb{Z}^d$, then $I_S(y)$ is contained in the coordinatewise box

$$-y_i \le x_i \le y_i$$

when all $y_i \ge 0$, and is empty otherwise. Thus all symmetric fibres are finite. The density is

$$\frac{|\mathbb{N}_0^d \cap [-N,N]^d|}{2(2N+1)^d} = \frac{(N+1)^d}{2(2N+1)^d} \to 2^{-(d+1)}.$$

□

8.2. **Parity reduction in the infinite dihedral group.** Let $D_\infty = \mathbb{Z} \rtimes C_2$. For $S_0, S_1 \subseteq \mathbb{Z}$ put

$$A = (S_0 \times \{0\}) \cup (S_1 \times \{1\}) \subseteq D_\infty.$$

Theorem 8.3 (Parity reduction in D_∞)**.** *The set A contains $BC \cup CB$ for some infinite $B, C \subseteq D_\infty$ if and only if at least one of the following holds:*

(i) there are infinite $X, Y \subseteq \mathbb{Z}$ with $X + Y \subseteq S_0$;
(ii) there are infinite $X, Y \subseteq \mathbb{Z}$ with $(X - Y) \cup (Y - X) \subseteq S_0$;
(iii) there are infinite $X, Y \subseteq \mathbb{Z}$ with $(X + Y) \cup (Y - X) \subseteq S_1$.

Proof. Suppose first that $BC \cup CB \subseteq A$. By the infinite pigeonhole principle, pass to infinite subsets of B and C each lying in a single parity layer.

If both lie in the rotation layer, write $B = X \times \{0\}$ and $C = Y \times \{0\}$. Then

$$(x,0)(y,0) = (x+y,0) = (y,0)(x,0),$$

so $X + Y \subseteq S_0$.

If both lie in the reflection layer, write $B = X \times \{1\}$ and $C = Y \times \{1\}$. Then

$$(x,1)(y,1) = (x-y,0), \qquad (y,1)(x,1) = (y-x,0),$$

so $(X - Y) \cup (Y - X) \subseteq S_0$.

Finally suppose one set lies in the rotation layer and the other in the reflection layer. Naming the rotation coordinates X and the reflection coordinates Y, the two products are

$$(x,0)(y,1) = (x+y,1), \qquad (y,1)(x,0) = (y-x,1),$$

so $(X + Y) \cup (Y - X) \subseteq S_1$.

Conversely, if any of (i), (ii), or (iii) holds, the same displayed product formulas show how to choose B and C in the corresponding layers so that $BC \cup CB \subseteq A$. □

Example 8.4. Taking $S_0 = \varnothing$ and $S_1 = \mathbb{N}_0$ gives the order-cone counterexample in D_∞. Alternatives (i) and (ii) are impossible. Alternative (iii) would require infinite $X, Y \subseteq \mathbb{Z}$ with

$$X + Y \subseteq \mathbb{N}_0, \qquad Y - X \subseteq \mathbb{N}_0.$$

Fixing $y \in Y$ would force $X \subseteq [-y, y]$, a finite set.

8.3. **A conjugation-invariant positive contrast.**

Theorem 8.5 (Conjugation-invariant sets)**.** *Let G be a countably infinite amenable group, let Φ be a two-sided Følner sequence, and let $A \subseteq G$ satisfy $\overline{d}_\Phi(A) > 0$. If A is conjugation-invariant, meaning $gAg^{-1} = A$ for every $g \in G$, then there exist infinite $B, C \subseteq G$ such that*

$$BC \cup CB \subseteq A.$$

Proof. By the product-set theorem of Moreira, Richter and Robertson [16, Theorem 1.3], applied in amenable groups, there are infinite $B, C \subseteq G$ with $BC \subseteq A$. Fix $b \in B$ and $c \in C$. Since $bc \in A$ and A is conjugation-invariant,

$$c(bc)c^{-1} = cb$$

belongs to A. Thus $CB \subseteq A$ as well. $\square$

9. Proof of Theorem 9.5

This section is not needed for the counterexamples, but it isolates a sharp one-dimensional threshold problem.

For a countably infinite amenable group G, define

$$\mathrm{d}^*_G(A) = \sup_{\Phi} \limsup_{N\to\infty} \frac{|A \cap \Phi_N|}{|\Phi_N|},$$

where the supremum is over all two-sided Følner sequences in G. Define

$$\rho(G) = \sup\{\mathrm{d}^*_G(A) : A \subseteq G \text{ contains no } BC \cup CB \text{ with } B, C \text{ infinite}\}.$$

Proposition 9.1. *If G is countably infinite, abelian and amenable, then $\rho(G) = 0$.*

Proof. If A has positive upper Banach density, the product-set theorem gives infinite $B, C \subseteq G$ with $BC \subseteq A$. Since G is abelian, $BC = CB$, hence $BC \cup CB \subseteq A$. $\square$

Lemma 9.2 (Finite-index cosets along Følner sequences)**.** *Let H be a subgroup of finite index m in a countable group G. If (Φ_N) is a right Følner sequence in G, then for every right coset Hg,*

$$\frac{|\Phi_N \cap Hg|}{|\Phi_N|} \to \frac{1}{m}.$$

If (Φ_N) is two-sided Følner, then $(\Phi_N \cap H)$ is a two-sided Følner sequence in H after deleting finitely many empty terms.

Proof. The right action of G on the finite set $H\backslash G$ is transitive. Let

$$p_N(Hg) = \frac{|\Phi_N \cap Hg|}{|\Phi_N|}.$$

The right Følner property implies that for each fixed $a \in G$,

$$\sum_{Hg\in H\backslash G} |p_N(Hga) - p_N(Hg)| \to 0.$$

Every limit point of the probability vectors p_N is therefore invariant under the transitive action of G on $H\backslash G$, and hence is the uniform probability vector.

If $h \in H$, then

$$|((\Phi_N \cap H)h) \triangle (\Phi_N \cap H)| \le |\Phi_N h \triangle \Phi_N|,$$

and similarly on the left. Since $|\Phi_N \cap H|/|\Phi_N| \to 1/m$, the Følner property passes to $\Phi_N \cap H$ inside H. $\square$

Proposition 9.3. *Let G be a countably infinite amenable group and suppose that $H \le G$ is an abelian subgroup of finite index m. Then*

$$\rho(G) \le 1 - \frac{1}{m}.$$

In particular, $\rho(D_\infty) \le 1/2$.

Proof. Let $A \subseteq G$ satisfy $\mathrm{d}^*_G(A) > 1-1/m$. Choose a two-sided Følner sequence Φ_N and $\eta > 0$ such that along a subsequence

$$\frac{|A \cap \Phi_N|}{|\Phi_N|} > 1 - \frac{1}{m} + \eta.$$

By Lemma 9.2,

$$\frac{|H \cap \Phi_N|}{|\Phi_N|} \to \frac{1}{m}.$$

Therefore, along the same subsequence,

$$|A \cap H \cap \Phi_N| \geq |H \cap \Phi_N| - |A^c \cap \Phi_N| \geq (\eta - o(1))|\Phi_N|.$$

Relative to the Følner sequence $\Phi_N \cap H$ in H, the set $A \cap H$ has positive upper density. Since H is abelian, the product-set theorem gives infinite $B, C \subseteq H$ with

$$BC \subseteq A \cap H.$$

Then $BC = CB$, so $BC \cup CB \subseteq A$. □

A sequence (F_N) of finite subsets of $\mathbb{Z}$ is called *symmetric Følner* if it is Følner in $\mathbb{Z}$ and

$$\frac{|F_N \triangle (-F_N)|}{|F_N|} \to 0.$$

For $S \subseteq \mathbb{Z}$ define

$$\mathrm{d}^*_{\mathrm{sym}}(S) = \sup_F \limsup_{N\to\infty} \frac{|S \cap F_N|}{|F_N|},$$

where the supremum is over symmetric Følner sequences. Define

$$\rho_{\pm}(\mathbb{Z}) = \sup\{\mathrm{d}^*_{\mathrm{sym}}(S) : S \subseteq \mathbb{Z} \text{ contains no infinite } X, Y \subseteq \mathbb{Z}$$
$$\text{with } (X+Y) \cup (Y-X) \subseteq S\}.$$

Lemma 9.4 (Layer analysis in D_∞)**.** *Let (Φ_N) be a two-sided Følner sequence in D_∞. Write*

$$F_N^{(0)} = \{n \in \mathbb{Z} : (n,0) \in \Phi_N\}, \qquad F_N^{(1)} = \{n \in \mathbb{Z} : (n,1) \in \Phi_N\}.$$

Then

$$|F_N^{(0)} \triangle F_N^{(1)}| = o(|\Phi_N|), \qquad |F_N^{(i)}| = \frac{1}{2}|\Phi_N| + o(|\Phi_N|),$$

both $(F_N^{(0)})$ and $(F_N^{(1)})$ are Følner sequences in $\mathbb{Z}$, and

$$|F_N^{(1)} \triangle (-F_N^{(1)})| = o(|F_N^{(1)}|).$$

Proof. Right multiplication by $(0,1)$ exchanges the two layers without changing the $\mathbb{Z}$-coordinate. The right Følner property therefore gives

$$|F_N^{(0)} \triangle F_N^{(1)}| = o(|\Phi_N|),$$

which implies the two layer-size estimates.

Right multiplication by $(u,0)$ sends

$$(n,0) \mapsto (n+u,0), \qquad (n,1) \mapsto (n-u,1).$$

Thus the right Følner property gives, for each fixed $u \in \mathbb{Z}$,

$$|(F_N^{(0)} + u) \triangle F_N^{(0)}| = o(|\Phi_N|),$$

and similarly for $F_N^{(1)}$. Since $|F_N^{(i)}| \sim |\Phi_N|/2$, both layer sequences are Følner in $\mathbb{Z}$.

Left multiplication by $(0,1)$ sends

$$(n,0) \mapsto (-n,1), \qquad (n,1) \mapsto (-n,0).$$

The left Følner property gives

$$|F_N^{(1)} \triangle (-F_N^{(0)})| = o(|\Phi_N|).$$

Combining this with $|F_N^{(0)} \triangle F_N^{(1)}| = o(|\Phi_N|)$ yields the claimed asymptotic symmetry of $F_N^{(1)}$. □

Theorem 9.5 (Threshold reduction for D_∞)**.** *One has*

$$\rho(D_\infty) = \frac{1}{2}\rho_\pm(\mathbb{Z}).$$

Consequently

$$\frac{1}{4} \leq \rho(D_\infty) \leq \frac{1}{2}.$$

Proof. First let $S \subseteq \mathbb{Z}$ contain no infinite $X, Y \subseteq \mathbb{Z}$ satisfying

$$(X+Y) \cup (Y-X) \subseteq S.$$

Put $A = S \times \{1\} \subseteq D_\infty$. By Theorem 8.3, A contains no symmetric product rectangle. If (F_N) is any symmetric Følner sequence in $\mathbb{Z}$, then

$$\Phi_N = F_N \times \{0,1\}$$

is a two-sided Følner sequence in D_∞, and

$$\frac{|A \cap \Phi_N|}{|\Phi_N|} = \frac{|S \cap F_N|}{2|F_N|}.$$

Taking suprema gives $\rho(D_\infty) \geq \frac{1}{2}\rho_\pm(\mathbb{Z})$.

Conversely, let

$$A = (S_0 \times \{0\}) \cup (S_1 \times \{1\}) \subseteq D_\infty$$

contain no symmetric product rectangle. By Theorem 8.3, there are no infinite $X, Y \subseteq \mathbb{Z}$ with $X + Y \subseteq S_0$. Hence S_0 has zero upper Banach density in $\mathbb{Z}$, for otherwise the product-set theorem in the amenable group $\mathbb{Z}$ would give such X, Y.

Again by Theorem 8.3, the set S_1 contains no infinite $X, Y \subseteq \mathbb{Z}$ with

$$(X+Y) \cup (Y-X) \subseteq S_1.$$

Let (Φ_N) be any two-sided Følner sequence in D_∞ and define $F_N^{(0)}, F_N^{(1)}$ as in Lemma 9.4. Then

$$\frac{|A \cap \Phi_N|}{|\Phi_N|} = \frac{|S_0 \cap F_N^{(0)}|}{|\Phi_N|} + \frac{|S_1 \cap F_N^{(1)}|}{|\Phi_N|}.$$

The first term has limsup zero because S_0 has zero upper Banach density and $(F_N^{(0)})$ is Følner in $\mathbb{Z}$. By Lemma 9.4, $(F_N^{(1)})$ is a symmetric Følner sequence in the sense used to define $\mathrm{d}^*_{\mathrm{sym}}$: it is Følner in $\mathbb{Z}$ and satisfies

$$|F_N^{(1)} \triangle (-F_N^{(1)})| = o(|F_N^{(1)}|).$$

Also $|F_N^{(1)}|/|\Phi_N| \to 1/2$, so the limsup of the second term is at most $\frac{1}{2}\rho_\pm(\mathbb{Z})$. Since (Φ_N) was arbitrary, $\mathrm{d}^*_{D_\infty}(A) \leq \frac{1}{2}\rho_\pm(\mathbb{Z})$. □

Proof of the final bounds. The upper bound $\rho(D_\infty) \leq 1/2$ follows from Proposition 9.3, since $\mathbb{Z} \times \{0\}$ has index two. For the lower bound, take $S = \mathbb{N}_0$. Then $\mathrm{d}^*_{\mathrm{sym}}(S) = 1/2$: every symmetric Følner sequence has asymptotically equal intersections with the two half-lines, while the intervals $[-N, N]$ attain the value $1/2$. Example 8.4 shows that S contains no infinite X, Y with $(X+Y) \cup (Y-X) \subseteq S$. Hence $\rho_\pm(\mathbb{Z}) \geq 1/2$, and Theorem 9.5 gives $\rho(D_\infty) \geq 1/4$. □

10. Finite patterns and final remarks

The obstructions in this paper are genuinely infinitary. Finite versions of the forbidden configurations often exist in abundance.

Proposition 10.1 (Finite symmetric product rectangles in the order cone)**.** *Let*

$$A = \mathbb{N}_0 \times \{1\} \subseteq D_\infty.$$

For every $r, s \in \mathbb{N}$ there are finite $B, C \subseteq D_\infty$ with $|B| = r$, $|C| = s$, and

$$BC \cup CB \subseteq A.$$

However, no such inclusion is possible with B, C both infinite.

Proof. Let

$$X = \{1, 2, \ldots, r\}, \qquad Y = \{r, r+1, \ldots, r+s-1\}.$$

Put $B = X \times \{0\}$ and $C = Y \times \{1\}$. If $b = (x, 0)$ and $c = (y, 1)$, then

$$bc = (x+y, 1), \qquad cb = (y-x, 1).$$

Since $x + y \geq 0$ and $y - x \geq 0$, both products lie in A. The infinite nonexistence follows from Theorem 8.1 with $S = \mathbb{N}_0$. □

Remark 10.2 (Related notes)**.** The logarithmic colouring counterexample is included here as the self-contained resolution of [14, Question 3.20]. The small diagonal-image obstruction for nonlinear ordered sumsets is not included as a main result, since it overlaps with Ackelsberg's negative results for the corresponding KMRR questions [1]. Keeping that material separate avoids priority ambiguity.

Acknowledgements

H.L. was supported by the National Natural Science Foundation of China (12501487), by China Scholarship Council and IBS-R029-C4. X.L. was supported by the Excellent Young Talents Program (Overseas) of the National Natural Science Foundation of China.

Declaration on the use of AI

The authors used generative AI tools to assist in discussing proof strategies, checking proofs, and improving exposition. All mathematical arguments, results, and conclusions were reviewed and verified by the authors.

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